Rulebook Project Requirements

Product requirements and step by step delivery flow

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Audience Product, engineering, design and quality assurance

Purpose

Rulebook must help a team understand what a software document says, what remains uncertain, who can do what, and what can be planned next. Every material interpretation must remain traceable to its source. Reliable understanding comes before generated maps, UI previews or delivery promises.

The target application has three primary screens: Project List, Doc Upload and Analyze. Its main product outputs are a branching role map and a UI Playground for documented frontend behavior. Planning and delivery forecasting build on a reviewed scope baseline.

Release position

The current private Vercel release is a session-based PDF reader with purpose and overview, dynamic role understanding, source-linked facts and scope analysis. Project persistence, editable approvals, branching maps, UI Playground and delivery forecasting are target capabilities. Preserved historical backend code is not evidence that those workflows are active in this release.

How to use this document

Use the contents page to jump to each section. Sections 2 to 5 establish scope and user flow; sections 6 to 20 define product and engineering requirements. Sections 21 to 23 define quality checks. Sections 24 to 27 contain the roadmap, decisions and handoff checklist; section 28 records the current API contract.

Requirement IDs in this document identify the product contract. RB ticket IDs retain the original nine-group roadmap. Live status refers to the protected release verified on 30 October 2026. Planned requirements are targets, not descriptions of delivered behavior.

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2 Product scope and release status

The immediate product must remain focused. Project List, Doc Upload and Analyze contain the workflow; maps, previews and planning appear within Analyze rather than becoming a collection of separate applications.

Capability	Current release	Target requirement
PDF source comparison	Live	Keep original pages, extracted text and warnings together.
Purpose and short overview	Live	Show module, current situation and intended work; preserve unknowns.
Roles and access interpretation	Live and bounded	Discover arbitrary labels and keep conditions, denials and source links.
Behaviors dependencies numbers	Live Group 3	Separate facts and preserve their meaning and provenance.
Scope boundaries and areas	Live Group 4	Keep work status and delivery scope independent.
Projects and saved documents	Not active in reader	Restore a secure project and version lifecycle.
Human correction and approval	Planned	Persist decisions and distinguish them from AI review.
Branching role map	Planned	Show role to feature to action branches with scope.
UI Playground	Planned	Simulate only documented frontend behavior.
Tasks effort and forecast	Planned	Use approved scope and explicit assumptions.
GitHub integration	Deferred to V3	Add optional code comparison after the document workflow is reliable.

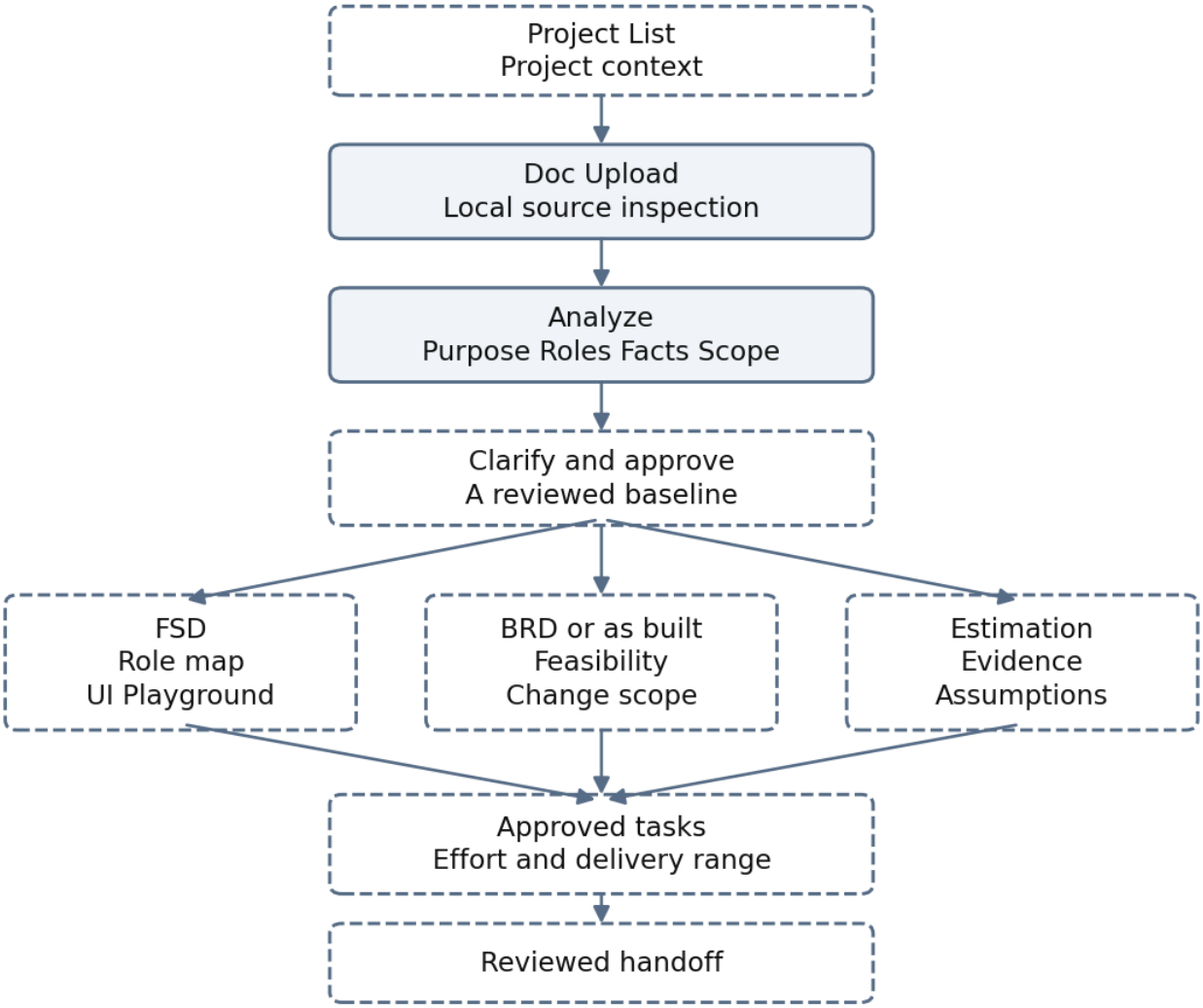
Scope exclusions for the near term

Automatic production code deployment, production permission provisioning, a universal generated backend, repository-based analysis and arbitrary OCR are outside the current document-understanding release. Additional source formats, organization administration and billing need their own release decisions.

Application access roles and roles found inside uploaded documents are separate concepts. A source document mentioning an Administrator never grants the uploader administration rights in Rulebook.

3 End to end product journey

The diagram defines the target journey. Solid nodes represent live capabilities inside the current single reader screen. Dashed nodes represent planned product workflow. Project context and review approval must be added before this becomes a persistent enterprise process.



Target user journey with purpose-specific outputs. Dashed nodes are planned capabilities.

A draft role map may be available before approval if unresolved branches remain visible. Approved planning, a formal handoff and delivery forecasts must refer to a named, reviewed baseline. A change to that baseline must invalidate affected downstream outputs.

4 Step by step project and document intake

W01 Create or reopen a project

Target flow. Preconditions: the user is authenticated, has access to the workspace, and has permission to create or open the project.

W01 Step 1 Open Project List. Show only projects the user may access, with name, purpose, latest document version and analysis status.

W01 Step 2 Select New project. Enter a project name and optional short business objective. Keep technology, repository and estimation fields optional.

W01 Step 3 Create the project and open Doc Upload. Assign a stable project ID and bind all later documents and analysis runs to it.

W01 Step 4 For an existing project, open its latest reviewed analysis or select Add document. Never combine unrelated example projects automatically.

W02 Open and inspect a document

W02 Step 1 Choose or drop a PDF. Validate format and size before reading. The current reader supports one local PDF per session.

W02 Step 2 Read every page in order and retain blank-page entries. Show progress, cancellation and useful errors.

W02 Step 3 Display the original page next to the extracted text. Preserve page references and table row and column relationships where reliable.

W02 Step 4 Inspect warnings and missing content. A title-only document cannot establish application requirements. A scanned page must not be treated as an empty specification.

W02 Step 5 Choose the processing policy and AI consent. If external AI is prohibited, keep the local reader available and do not transmit document content.

W02 Step 6 For the persistent product, attach the document to the project as a new version and apply the configured retention policy. Do not silently save a local-only document.

Exit condition: the user can inspect the source and knows whether it is readable, which version is being analyzed, and whether external processing is permitted. Project storage and the configurable processing-policy gate remain planned; the current reader keeps its session in browser memory.

5 Step by step analysis and recovery

The current Understand document action runs the first two processing phases below. Formal human confirmation and baseline approval are planned.

W03 Step 1 Check that readable content is sufficient and within the analysis limits. Reject oversized input clearly rather than analyzing an undisclosed subset.

W03 Step 2 Require explicit consent naming Google Gemini. Send extracted page text and validated table metadata to the protected server; keep the original PDF local.

W03 Step 3 Identify the primary document purpose and a short understanding of module, current state and intended work. Keep unsupported fields unknown.

W03 Step 4 Discover source roles, inventory requirements and permission cells, interpret assignments, and run a separate review. Distinguish performer, requester, approver, viewer and subject.

W03 Step 5 Extract atomic human and automatic behaviors, technical dependencies and numeric candidates. Preserve original wording, units, referents, conditions and denials.

W03 Step 6 Revalidate source identity in the browser and show roles and facts immediately. Keep omitted or rejected source items visible as unresolved.

W03 Step 7 Run scope analysis as a separate request. Discover functional areas, classify work status and delivery scope independently, and review each axis against source evidence.

W03 Step 8 Revalidate scope in the browser. Show existing, proposed, included, excluded, undecided and unspecified states, with independent filters and page links.

W03 Step 9 If scope fails or is stopped, preserve the earlier results. Offer Retry scope only. Prevent a stopped, replaced or stale response from overwriting the current document.

W03 Step 10 In the target review workflow, correct understanding, resolve questions and approve a versioned baseline before generating committed delivery outputs.

Exit conditions

A successful AI run produces a reviewable draft, not proof of implemented software or complete interpretation. An incomplete run must identify the affected passages and preserve every source item in its managed inventory. A health response that says AI is configured is not evidence that a particular document was interpreted correctly.

6 Requirements for the three screens

Project List

SCR-01 Show project name, latest source version, last analysis status and unresolved-review count. Search and open projects without exposing another workspace's data. This screen is planned in the focused Vercel product.

SCR-02 Provide create, rename and delete actions according to permissions. Deletion requires a warning describing the affected records and must follow the retention and recovery policy.

Doc Upload

SCR-03 Use one clear upload action with drag and drop. Show supported formats and limits before selection. Keep the local processing policy visible.

SCR-04 After reading, show original and extracted views, page navigation, reading progress, warnings, Open original, replace and clear actions. Reset consent and analysis for every replacement, even if the filename is unchanged.

Analyze

SCR-05 Start with a short understanding and purpose. Provide source-linked roles, facts and scope as the live foundation. Advanced outputs remain organized inside this screen.

SCR-06 Target sections are Source, Understanding, Roles, Facts, Scope, Questions, Review, Role Map, UI Playground, Tasks and Forecast. Show only sections relevant to document purpose and implemented capabilities.

SCR-07 Keep a selected item's source passage one action away. Selecting an item must open the correct document version and page, not only a generic source list.

SCR-08 Show empty, busy, partial, failed, stopped and completed states clearly. Preserve usable results when a later stage fails. Do not use a single success badge to imply every stage succeeded.

Navigation principle

The reader must always know the project, document version and analysis stage. Functional areas and role branches should reduce scanning. Avoid long undifferentiated cards, repeated forms and paragraphs that simply reproduce the whole FSD.

7 Project persistence and document versions

These requirements define the persistent product. The current reader does not save its PDF, extracted text or results, and reloading clears its session. Historical database and file-storage integrations remain preserved on their original host; they are not automatically part of the Vercel reader.

PRJ-01 Create stable workspace, project, document and version IDs. Bind every analysis result to the exact source version and content fingerprint.

PRJ-02 Support multiple related documents in one project only after source identity and version handling are available. Show document name, purpose, version and status separately.

PRJ-03 Treat replacement as a new document version. Preserve prior versions and decisions according to policy; never overwrite source history silently.

PRJ-04 Let an authorized user rename or delete a project or document through an explicit action. Show the effect on linked analysis, decisions and handoff artifacts.

PRJ-05 Persist reviewed findings, clarification answers and approvals with actor and time. Store user statements separately from original document claims.

PRJ-06 On revision, mark affected findings and downstream artifacts as needing review. Unaffected items may retain history only when their provenance can be demonstrated.

PRJ-07 Provide optimistic concurrency or equivalent protection so one reviewer cannot silently overwrite another reviewer's decision.

Version change flow

W04 Step 1 Add the replacement source and create a new version.

W04 Step 2 Rebuild its source inventory and compare affected statements and table cells.

W04 Step 3 Flag changed, removed and new findings for review.

W04 Step 4 Invalidate affected map branches, previews, tasks, acceptance checks, effort and forecasts.

W04 Step 5 Reapprove the affected baseline before exporting a new committed handoff.

Acceptance: opening a saved project must restore the correct version and decisions. It must never carry an old approval or reported test pass into changed requirements without explicit revalidation.

8 Source extraction and purpose requirements

DOC-01 Keep original page boundaries, page numbers and reading order. Retain blank pages and distinguish extraction failure from absent requirements.

DOC-02 Preserve headings, identifiers, clauses, signs, percentages, dates, times and original table relationships. Reconstructed data must retain the original source lines.

DOC-03 When geometry is ambiguous, flag the region and keep it reviewable. Absence of a warning does not establish complete extraction. OCR and arbitrary table layouts are not current guarantees.

DOC-04 Maintain an independent source inventory before model interpretation. Count source units from the input, not from whichever citations the model returns.

PUR-01 Recognize requirements specifications, business requirements, existing-system descriptions, feature inventories and estimation documents. Mixed purpose requires separate supporting passages; otherwise remain unclear.

PUR-02 Explain the module, current situation and intended work in a short statement. Each established field needs supporting source evidence. Unknown means not established, not proof that the source contains no information.

PUR-03 Allow the user to correct and confirm understanding in the target review workflow. Preserve the initial AI suggestion and mark the correction as user-provided information.

Purpose routing rules

An estimation report that lists features remains an estimation report unless a separate primary purpose is supported. An as-built desktop specification must not acquire invented HTTP endpoints, JWT scopes or browser roles. A title or filename alone cannot establish a product's behavior.

Acceptance checks

- A blank or title-only source produces no invented requirements.
- Every accepted passage opens at its verified page and version.
- A source limit is visible before an incomplete interpretation could be presented as complete.

9 Dynamic roles and access understanding

Roles must be discovered from the document, whatever their names. Office Admin, ITEngineer and Electronics Engineer are examples of possible source labels, not a supported-role dictionary. Operational caps on a single run are separate from the vocabulary of roles the product can recognize.

ROL-01 Preserve original role names and definitions. Use an alias or broader category only when supported, and retain the evidence. Do not force every administrator-like title into an invented canonical role.

ROL-02 Keep distinct roles distinct. A broad group such as Staff is not automatically identical to Teacher or Engineer. A referenced person record is not automatically an application user.

ROL-03 For each role-linked behavior, retain action, resource, effect, scope, conditions, exceptions and source. Distinguish allow, deny, require and not established.

ROL-04 Distinguish performer, requester, approver, viewer and subject. A person requesting an override does not acquire approval authority.

ROL-05 Preserve own, assigned, class-specific, team-wide and global scope. Never expand assigned-only access to all records.

ROL-06 Respect the actual row and role column of an access matrix. No Access must not become allowed behavior. Do not interpret unrelated table headers as roles.

ROL-07 Keep automatic system behavior separate from human ownership. If ownership is unsupported, retain the requirement as unresolved rather than dropping it.

Required access record

Field	Meaning
Role and relationship	The source actor and whether they perform, request, approve, view or are a subject.
Action and resource	The operation and the object it concerns.
Effect and record scope	Permission, prohibition or obligation, including own or assigned boundaries.
Conditions and provenance	Trigger, approval, timing, exception, source version, page and passage.

Acceptance: adversarial fixtures must not reverse denials, swap matrix columns, merge unsupported aliases, expand record scope or turn an approver into a requester.

10 Behaviors dependencies and numeric facts

Group 3 creates facts that can be checked against source. A quotation establishes provenance; a separate review checks interpretation. AI-reviewed is distinct from human-approved.

FACT-01 Split compound behaviors into separate actions while retaining their parent statement. Create, update, deactivate and assign are separate actions. Preserve negation, conditions and exceptions on each affected action.

FACT-02 Include automatic operations, scheduled locks, procedure calls and state changes. A dependency or time fact must not replace the corresponding behavior record.

FACT-03 Discover documented technologies, databases, procedures, external services and tools. Preserve their full source relationship instead of allowing a rewritten phrase to omit conditions.

FACT-04 Inventory numeric candidates independently of model output. Preserve exact source values, units, referents and qualifiers without conversion, rounding or invented calculations.

FACT-05 Separate measurements, clock times, dates, versions, identifiers and headings. A clock cutoff is not an effort duration; a software version is not a work estimate.

FACT-06 Distinguish baseline and revised figures. A percentage needs its denominator or referent; LOC, hours, file count and business days must not be mixed.

Source example	Required interpretation
1,200 LOC and 16 hours	Two values with distinct units and the same stated baseline.
900 LOC and 12 hours revised	Two revised values, not replacements for the original evidence.
11:59 PM	Clock time with its full time designator and documented referent.
Three features	A written count associated with features.
Runtime version 3.13	Version identifier; it cannot contribute to effort totals.

All examples are illustrative. Missing or rejected interpretations remain visible. Facts with unstated units or referents must show Not established rather than a guessed value.

11 Scope boundaries and functional areas

Group 4 has two independent classification axes. Work status records whether behavior exists or is proposed. Delivery scope records whether work is included, excluded, undecided or not established.

Document statement	Work status	Delivery scope
An existing export stays in this release	Existing	Included
A proposed biometric feature is excluded	Proposed	Excluded
Offline entry remains undecided	Not established	Undecided
A role cannot delete archived records	Not established unless separately supported	Not established
A source is silent on a feature's scope	Not established unless separately supported	Not established

SCP-01 Separate mixed clauses when they describe different work statuses or scope decisions. Preserve the conditions attached to each clause.

SCP-02 Require independent source support for each positive axis. An exclusion or undecided decision alone cannot establish proposed work. A permission restriction is not a project exclusion.

SCP-03 Discover functional areas dynamically, including excluded and undecided features. Group related current and proposed work by capability, not only by the implementation medium.

SCP-04 Compile group evidence from verified source-unit IDs. Mark labels that are not used in the document as suggested organizing labels.

SCP-05 Review work status, delivery boundary and area membership separately. Clear an unsupported axis without discarding another supported axis. Bind review decisions to explicit item IDs.

SCP-06 Provide independent work, scope and area filters, source-page links and a complete source coverage view. Keep unresolved items unclassified and reviewable.

Scope is an interpretation of the document, not evidence of a signed delivery commitment. Human baseline approval is a later product requirement.

12 Missing information conflicts and questions

Group 5 turns gaps into focused review work. This workflow is planned. The current unresolved states and extraction warnings provide inputs, but are not a complete conflict detector or clarification system.

CLR-01 Detect references to missing specifications, rule definitions, procedure contracts, source documents or declared feature sections. Show what is missing and the passage that refers to it.

CLR-02 Present potential conflicts with both passages, their versions and their affected facts. Do not silently select one statement as the truth.

CLR-03 Distinguish an actual contradiction from an exception, a different role scope or a documented version change. Keep uncertain conflict detection reviewable.

CLR-04 Generate one focused question for each material uncertainty. State why it matters and which role, behavior, UI state, task or estimate is affected.

CLR-05 Classify whether a gap blocks interpretation, implementation, acceptance or estimation. Do not block unrelated reviewed work automatically.

CLR-06 Keep unanswered questions visible and exclude unsupported answers from planning. Do not fill an absent API contract from general model knowledge.

Question record

Field	Required content
Question and basis	The decision needed, referring passages and missing or conflicting information.
Impact	Affected source items, functional area and downstream outputs.
State	Open, answered, deferred or resolved through documented review.
Answer provenance	Who answered, when, and whether the answer is a user statement or new source evidence.

Example: a document refers to an archive procedure but omits its input contract. Ask for the contract and show the affected integration work. Do not invent parameter names, response fields or an endpoint URL.

13 Human review approval and revisions

Group 6 establishes the human decisions required for a committed handoff. It is planned. Current AI review labels must never be renamed Approved or Verified implementation.

W05 Step 1 Review the purpose and short understanding. Correct the module, as-is situation or requested work without overwriting the original AI proposal.

W05 Step 2 Review findings and choose approve, edit, reject or leave unresolved. Record the decision and its source version.

W05 Step 3 Answer clarification questions. Keep user-provided answers distinguishable from quotations in the uploaded source.

W05 Step 4 Approve only the included scope that is sufficiently established. Record unresolved or deferred work separately.

W05 Step 5 Create a named baseline with approved findings, scope decisions, answers, reviewer identity and time.

W05 Step 6 Generate formal maps, tasks, acceptance checks and forecasts against that baseline.

W05 Step 7 When a source changes, mark affected decisions and outputs stale and require a new review before publication.

REV-01 Rejected findings must remain in history but stay out of planning and forecast totals. Editing must retain the earlier value and the reason for change.

REV-02 A review decision must be scoped to a project and source version. The system must not reuse approval merely because another file has the same name.

REV-03 A user answer can resolve a question but does not become an original document quotation. Preserve its own provenance.

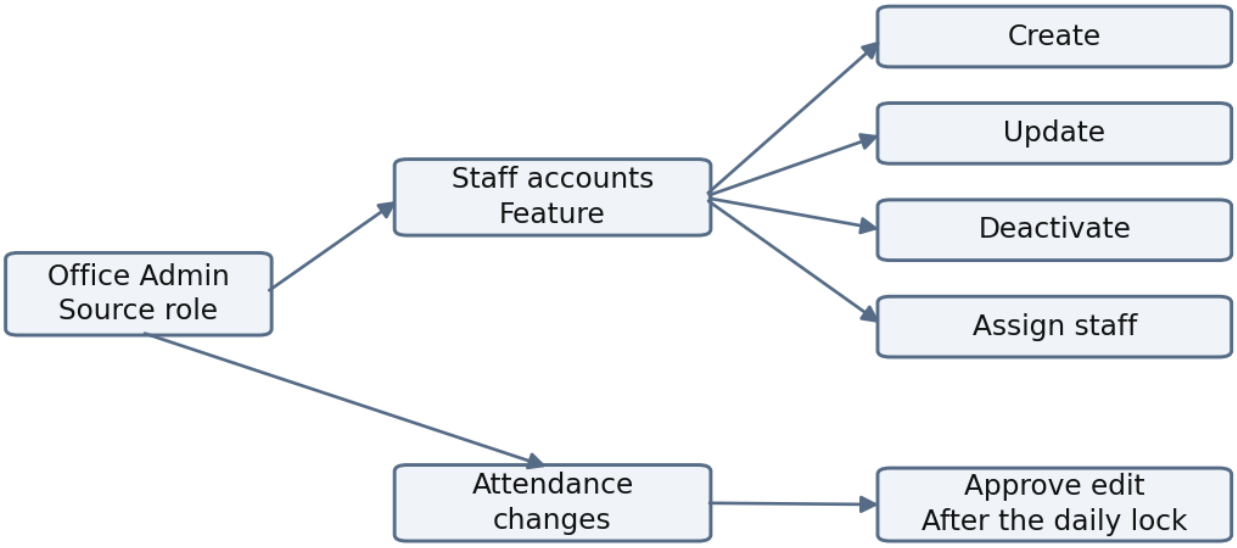
REV-04 Separate finding approval, baseline approval and reported implementation test results. They are different decisions with different evidence.

Acceptance

Reload must preserve authorized review decisions. A rejected requirement cannot appear in the generated backlog. Replacing a source must mark its affected approvals stale. An unresolved blocking question must remain visible in downstream work.

14 Branching role map and architecture layers

The role map is a planned core output. It must help users understand responsibilities faster than rereading the document. Render role to feature to action branches, with permission effect, record scope and source references attached to the relevant action.



Illustrative role branches only. Each action is separate; arrows show grouping, not execution order. These are example labels.

MAP-01 Provide collapse, expand, role and functional-area filtering. Keep unresolved ownership in a visible separate branch. Use shared references for shared actions rather than duplicating claims without traceability.

MAP-02 Clicking a node must reveal the original requirement, conditions, approval relationships and exact source version and page.

MAP-03 Separate documented frontend behavior from backend responsibilities. Do not create API endpoints, routes, database policies or JWT scopes when the source does not establish them.

Frontend view	Backend view
Screen and component visibility	Authorization and validation responsibilities
Navigation and route behavior	Documented API or procedure contracts
Forms and visible states	Data operations and integration boundaries
UI restrictions and messages	Scope enforcement and business rules

The map describes documented or approved requirements. Drawing a route guard or permission node does not implement that control in the source application.

15 UI Playground requirements

The UI Playground is planned and limited to the UI layer. It should demonstrate supported interactions and source-derived states without implying a generated production backend exists.

UI-01 Identify only documented user-facing screens, forms, controls, navigation, visible validations and states. Treat backend jobs, procedures and storage work as non-UI unless a user-facing requirement connects them.

UI-02 Generate a role-specific preview using isolated example data. Selecting a role changes the preview according to documented visibility and record scope.

UI-03 Support meaningful interactions such as form input, validation messages, list filtering, dialogs and navigation when the specification establishes them.

UI-04 Keep every preview element linked to a source requirement or an explicitly labeled design suggestion. A design suggestion must not silently become an approved requirement.

UI-05 Mark unsupported UI patterns and missing details clearly. A generic template must not be presented as a complete implementation of an arbitrary FSD.

UI-06 Run generated previews in an isolated environment with no production credentials, arbitrary network access or access to other projects' source data. Use mocked integrations.

Step by step playground flow

W06 Step 1 Open an eligible FSD analysis and choose UI Playground.

W06 Step 2 Select a documented role and feature. Show the governing source and unresolved UI details.

W06 Step 3 Generate or open a preview of the approved or explicitly labeled draft UI slice.

W06 Step 4 Interact with supported controls and switch documented states or roles.

W06 Step 5 Record design feedback or required clarification against its source item. Regenerate after approval or revision.

Acceptance: a role switch cannot display a documented forbidden action as allowed. Missing UI requirements must remain unknown. Preview interaction success is not an implementation test pass for the real application.

16 Outputs according to document purpose

Purpose controls emphasis and available outputs. The same dense layout must not be imposed on every document type. A mixed document may support multiple sections, each grounded in its own evidence.

Purpose	Primary output	Important boundary
FSD or detailed requirements	Roles, access, behaviors, scope, UI and technical requirements; draft maps and UI previews when eligible.	Only source-supported UI belongs in Playground.
BRD	Business objectives, stakeholders, scope, assumptions, feasibility questions and risks.	Do not invent technical contracts or detailed screens.
Existing system description	As-is capabilities, dependencies, defects and explicit change requests.	Current behavior is not automatically work to build.
Estimation or analysis report	Source estimates, units, benchmarks, assumptions and delivery risks.	Documented numbers are source claims, not newly validated forecasts.
Feature inventory	Feature groups, role links, unknown details and clarification needs.	A feature title alone does not define full behavior.
Mixed or unclear	Distinct supported purpose sections, or an unclear result with missing evidence.	Ask for confirmation; do not force an FSD label.

OUT-01 Let an authorized reviewer confirm or correct purpose. Preserve the previous classification and re-evaluate affected outputs.

OUT-02 A BRD or estimation document can produce a delivery forecast only after scope, effort, capacity and assumptions are sufficiently reviewed.

OUT-03 Unknown purpose or insufficient content must not trigger invented role maps, tasks or UI scenes. Keep the source comparison available.

The current release identifies purpose and shows a common foundation. Dedicated purpose-specific planning views belong to the later roadmap.

17 Tasks dependencies and acceptance checks

Group 7 is planned. Generate proposed tasks from approved included scope, rather than treating every source paragraph as new implementation work.

TASK-01 Each task must describe one reviewable outcome and link to the approved scope items and baseline that justify it.

TASK-02 Separate frontend, backend, integration, data, testing and deployment work when the source or a reviewed engineering decision supports the separation. Mark decomposition as a planning decision.

TASK-03 Record prerequisites, dependencies and blocking clarification questions. Detect circular dependencies and show them for review.

TASK-04 Keep unchanged existing behavior and excluded work out of build-task totals unless a reviewer explicitly adds regression or change work.

TASK-05 Draft observable acceptance checks. Distinguish a suggested check from an executable automated test and from a test that has actually been run.

Task field	Requirement
Identity and scope	Stable ID, outcome, baseline and source links.
Responsibility	Suggested engineering discipline; human owner only when assigned.
Prerequisites	Dependency IDs, blocking questions and integration needs.
Acceptance	Observable expected results and negative or boundary cases.
Effort	Reviewed range, unit, assumptions and estimation state.
Execution evidence	Not run by default; a reported pass needs evidence and source version.

Example acceptance checks for an approved create-account UI include required-field validation, authorized success, unauthorized rejection and the documented record scope. Endpoint names and validation rules must come from source or a recorded engineering decision.

18 Effort review and delivery forecasting

Groups 8 and 9 are planned. Keep documented estimate evidence, developer-entered effort, computed totals and delivery forecasts separate. LOC, feature count and benchmark labels alone do not establish elapsed delivery time.

W07 Step 1 Collect documented totals and benchmarks with units, referents, time periods and qualifiers.

W07 Step 2 Review approved tasks and identify work that is excluded, unchanged, blocked or not yet estimated.

W07 Step 3 Record task-level effort ranges, method, assumptions and reviewer. Explain any use of comparison projects or productivity rates.

W07 Step 4 Calculate totals from approved estimated work, avoid double counting and show unestimated work separately.

W07 Step 5 Enter team availability, productive hours, skills constraints and the working calendar.

W07 Step 6 Apply task dependencies, parallelism, capacity and explicit buffers to calculate a delivery range.

W07 Step 7 Show assumptions, remaining blockers and the effect of scope or capacity changes. Do not show a firm completion date when required inputs are absent.

EST-01 Every calculated value must show its inputs and method. A value from the document must remain labeled documented estimate evidence.

EST-02 Allow developers to revise effort ranges and assumptions. Preserve earlier estimates with their baseline and reviewer.

FRC-01 Expose team capacity and calendar inputs. Do not invent engineer availability, working days, holidays or a project start date.

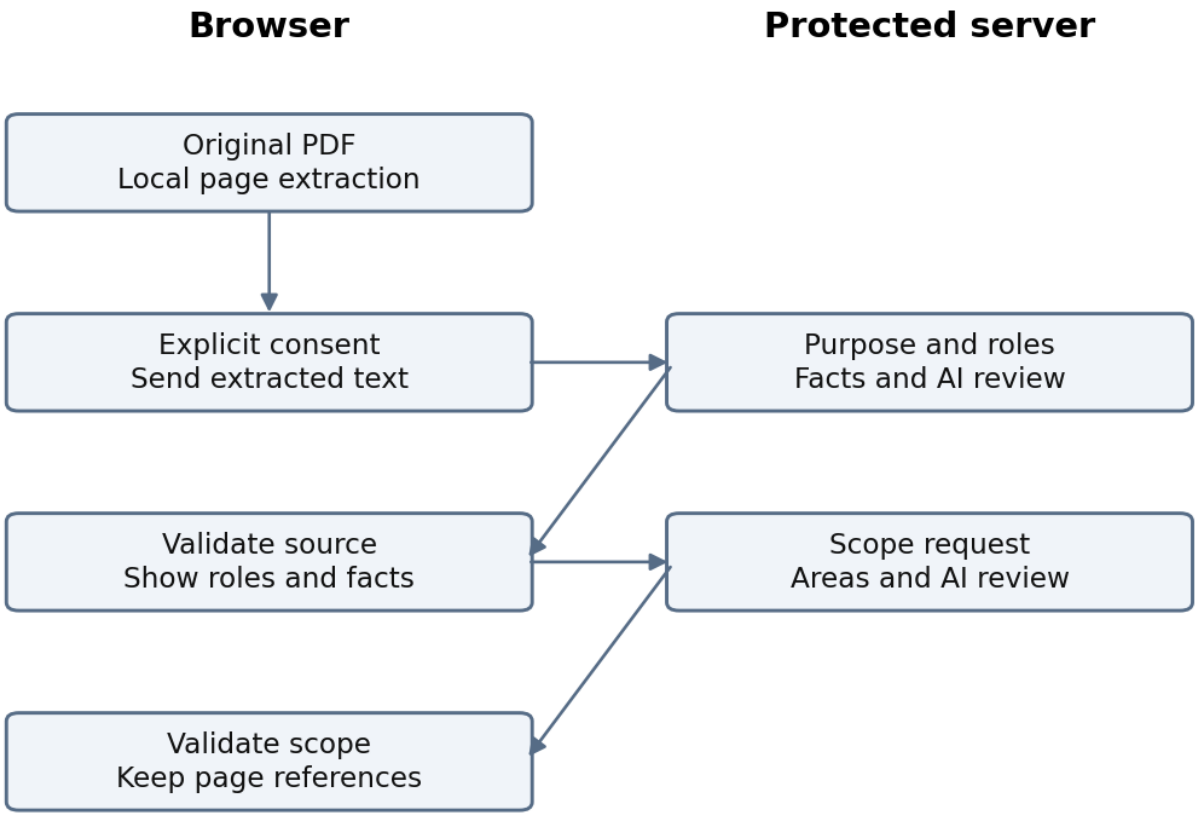
FRC-02 Produce an explainable range rather than a false precision score. A forecast meter must communicate what is known, what is assumed and what is blocked.

FRC-03 Changes in approved scope, dependencies, effort or capacity must mark the previous forecast stale and allow recalculation. Never silently reuse a date from an old baseline.

Acceptance: removing excluded work must remove its contribution from totals; adding a blocking dependency must affect the schedule; missing capacity must produce an incomplete forecast rather than a guessed date.

19 Architecture and core data relationships

The live architecture has browser-local reading and protected AI endpoints. Project storage and reviewed baselines are additions to this architecture, not a claim that legacy storage has already been migrated.



Current progressive analysis flow. Each browser result is revalidated before display.

Record	Essential relationships
Document version	Project, content fingerprint, page inventory and processing policy.
Analysis run and finding	Version, stage, model metadata, source units, evidence and review state.
Decision and baseline	Finding or question, actor, time, user answer and approved scope version.
Downstream artifact	Baseline, map or preview or task or estimate, assumptions and staleness.

In the target system, every task and forecast must trace through a baseline to decisions, findings and the exact source version. The model provider is an external processor, not the authoritative store of project truth.

20 APIs access and document privacy

Current server contract

Endpoint	Current role
POST /api/document-purpose	Consented purpose, role and fact analysis.
POST /api/document-scope	Consented independent scope and functional-area review.
GET on these endpoints	Safe health and version metadata; no model call or secret value.

SEC-01 Enforce access and source limits on the server. Keep credentials server-side and reject client-supplied keys, model overrides, unknown fields, cross-origin requests and spoofed hosts.

SEC-02 The private demo relies on Vercel Standard Protection and redirects public entry aliases to a protected immutable deployment. This is not a complete multi-tenant enterprise identity system.

SEC-03 For the persistent product, authorize every project, document, decision, artifact and export against workspace membership and action permissions. Frontend visibility must never substitute for backend authorization.

SEC-04 Model document text is untrusted input. Embedded instructions must not authorize tools, reveal credentials or override the application's processing policy.

Confidential documents

PRIV-01 Use an explicit workspace and document processing policy. If external processing is prohibited, allow local preview and block external AI. A general consent checkbox must not override an owner's confidentiality policy.

PRIV-02 Explain which content leaves the browser and which service receives it. Separate consent for AI processing from permission to store sources or share exports.

PRIV-03 The target persistent product must define retention, deletion, access revocation, encryption requirements and audit access before enterprise onboarding. Do not log raw source text, prompts, credentials or provider payloads.

The current reader has explicit AI opt-in and no app-level document persistence. A configurable policy gate, enterprise identity and retention workflow remain planned. Any data migration must be separately designed and authorized.

21 User experience and nonfunctional requirements

UX-01 Use a modern restrained layout, mild gradients and clear spacing. Use functional hierarchy and readable typography rather than many nested cards or a maze of forms.

UX-02 The main result must be easier to understand than the source document. Use branch maps for responsibility structure, tables for comparisons and disclosures for detailed evidence.

UX-03 Use color consistently for states, while retaining text labels. Do not communicate denied access, unresolved findings or exclusions by color alone.

UX-04 Support keyboard operation, visible focus, labeled controls, meaningful table headers and announced progress. Source navigation must work without relying on hover.

UX-05 Keep the interface usable on narrow screens. Preserve source and result context when views stack; allow deliberate horizontal scrolling for large tables.

NFR-01 Expose stage progress and allow the user to stop waiting. Long AI work must not freeze local reading or erase completed results.

NFR-02 Apply bounded input, output and retry policies. Every reached cap must be visible as a limitation or incomplete result; it must not masquerade as full coverage.

NFR-03 Measure document size, stage latency, provider calls, failure codes and source coverage without logging private content. Trace a run through a safe diagnostic ID.

NFR-04 Separate provider availability, application correctness and semantic quality metrics. A successful build or healthy endpoint does not establish interpretation accuracy.

NFR-05 Validate performance and concurrency against representative documents before setting commercial latency or availability commitments. No production SLA is established by the current demo.

NFR-06 Introduce per-workspace usage controls, cost visibility and an operational budget before a wider paid release. The private demo currently uses provider limits with the custom daily cap disabled.

22 Failure flows and operational limits

Condition	Required user outcome
Wrong format empty or corrupt file	Explain the input problem; make no AI call.
Password-protected source	Ask for an unlocked copy; do not bypass protection.
Scanned or unreadable pages	Show missing-text warnings; do not invent content.
Local preview limit exceeded	Reject clearly and suggest splitting the source.
AI source limit exceeded	Keep local reading available; do not silently truncate.
Provider timeout or rate limit	Show stage failure or partial results; preserve source and completed work.
Scope request fails	Keep roles and facts; offer scope-only retry.
Stop replace or clear	Prevent late responses from replacing current state; revoke local object URLs.
Invalid quote ID or assignment	Reject the affected claim and retain its source as unresolved.
Revision after approval	Mark affected decisions and downstream outputs stale.

Current limits as implementation facts

Boundary	Current value
Local PDF reader	20 MB, 150 pages, 1,000,000 extracted characters.
AI source payload	150 pages and 120,000 extracted characters; API body limit is also enforced.
Role understanding	160 inventoried source statements; up to 64 discovered roles per run.
Scope analysis	400 source passages and 24 functional areas; area-cap hits flag incompleteness.
Provider execution	35-second call timeout within a bounded request budget; function maximum 240 seconds.

These are current operational boundaries, not a promise of perfect extraction within them. New formats, larger documents, OCR and higher limits require explicit validation and product decisions.

23 Verification and release acceptance

Required test families

- Extraction: real PDF bytes, page boundaries, known table geometry, wrapped labels, blank pages, unsupported fonts and unreadable content.
- Meaning: arbitrary role labels, source-backed aliases, denied and assigned-only access, requester versus approver, and automatic operations.
- Facts and scope: compound actions, dependency provenance, numeric units and baselines, clock versus duration, mixed clauses, independent scope axes and undecided versus unspecified.
- Lifecycle: consent, protected access, stale-response isolation, scope-only retry, source replacement, review persistence and revision invalidation.
- Experience: actual authenticated browser upload, PDF rendering, source navigation, keyboard operation, responsive layout and purpose-specific output.

Measurable acceptance

QA-01 Every accepted claim must have source evidence or a distinct user-decision provenance. Every managed source item must have a visible disposition.

QA-02 Critical approved test fixtures must have zero denial reversals, matrix-column swaps, unsupported permission expansions or false scope commitments.

QA-03 A task or forecast must never include rejected or excluded work without an explicit recorded planning decision.

QA-04 Before enterprise release, verify workspace isolation, storage and retention controls, real browser flows and operational budgets. A private demo gate is not enterprise approval.

Current verification boundary

The 30 October release passed the regression suite, TypeScript, build and strict live synthetic-PDF checks. Scope validation reviewed 15 passages and produced 11 scope items in five areas. Dedicated Group 3 checks covered seven behaviors, four dependencies and ten numeric candidates. Role checks retained 16 requirements and 24 permission cells. These are bounded fixtures, not universal accuracy measurements.

Full private-document live model accuracy and authenticated browser upload/visual QA remain to be completed for broader release evidence. Do not describe AI-reviewed outputs as verified business software.

24 Roadmap for extraction and understanding

This table preserves the original nine-group RB roadmap. Titles and acceptance criteria identify the contract; historical repository filenames sometimes reuse an RB number for a narrower experiment.

Ticket	Requirement	Done when	Status
RB-01	Inspect extraction	Captured and lost text, tables and numbers are recorded for reviewed examples.	Complete
RB-02	Preserve page boundaries	Every extracted passage retains its original page.	Live
RB-03	Preview extracted content	Users compare app reading with the original and see warnings.	Live
RB-04	Identify purpose	Requirements, existing-system and estimation purposes have evidence; mixed and unclear are supported.	Live
RB-05	Short understanding	Module, current situation and intended work are concise; unknowns stay explicit.	Live
RB-06	Correct understanding	Users can edit and confirm purpose and understanding.	Planned
RB-07	Extract behaviors	Separate documented human and automatic behaviors retain source links.	Live
RB-08	Extract dependencies	Technologies, databases, procedures and tools have source evidence.	Live
RB-09	Extract numeric meaning	Exact values retain units, referents and qualifiers.	Live
RB-10	Separate existing and proposed	Current descriptions and requested changes remain distinct.	Live
RB-11	Capture scope boundaries	Included, excluded, undecided and unspecified work stay separate.	Live
RB-12	Group functional areas	Related work is grouped with source references and explicit label basis.	Live

Groups 1 to 4 provide the analysis foundation, with RB-06 still a review-workflow gap. Live status means implemented in the bounded current release, not complete support for all documents or production implementation of the analyzed application.

25 Roadmap for clarification review and planning

Groups 5 to 7 are planned. Complete the correction and approval path before presenting downstream planning as a committed delivery baseline.

Ticket	Requirement	Acceptance criterion
RB-13	Missing referenced information	Flag referenced rules, specifications or dependencies absent from the uploaded sources.
RB-14	Conflicting statements	Show both passages for each potential conflict and keep the decision reviewable.
RB-15	Focused questions	Each clarification question explains the uncertainty and the affected work.
RB-16	Approve edit or reject	Persist decisions and keep rejected findings out of planning.
RB-17	Clarification answers	Link answers to questions and identify them as user-provided information.
RB-18	Document revisions	Changed source content marks affected findings for another review.
RB-19	Small proposed tasks	Each task describes one reviewable outcome linked to approved scope.
RB-20	Dependencies and blockers	Tasks show prerequisites; unresolved questions remain visible.
RB-21	Acceptance checks	Each task has observable checks; AI suggestions await approval.

Recommended delivery order

M1 Step 1 Complete RB-06 and the Group 5 clarification model.

M1 Step 2 Add secure project persistence and the Group 6 review and revision lifecycle.

M1 Step 3 Build a narrow source-linked role map and a supported UI Playground slice using that baseline.

M1 Step 4 Add Group 7 task decomposition and acceptance drafting.

Map and Playground work serve the product's core promise. Their first release should demonstrate a small approved source slice end to end, with explicit unsupported states, rather than claiming universal generated applications.

26 Roadmap for estimation and forecasting

Groups 8 and 9 extend reviewed analysis into engineering planning. Their outputs depend on approved scope and explicit inputs; they must not make missing information look precise.

Ticket	Requirement	Acceptance criterion
RB-22	Documented estimate evidence	Document totals, benchmark figures and calculated values remain distinguishable.
RB-23	Task effort review	Developers enter or revise effort ranges and record assumptions.
RB-24	Explainable effort total	Include approved work, avoid double counting and identify unestimated tasks.
RB-25	Team capacity	Engineer availability and productive hours are editable and visible.
RB-26	Delivery range	Account for effort, dependencies, capacity and assumptions; flag missing inputs.
RB-27	Purpose-specific analysis	FSD views emphasize behavior and acceptance; estimation views emphasize effort, assumptions and delivery risks.

Milestones after the review foundation

M2 Step 1 Validate estimate-evidence extraction against representative reports and benchmark comparisons.

M2 Step 2 Add editable task ranges and total calculation with clear exclusions and unknowns.

M2 Step 3 Add capacity and calendar inputs, then explainable dependency-aware forecasts.

M2 Step 4 Release purpose-specific FSD, BRD and estimation views within Analyze.

M2 Step 5 Evaluate optional GitHub linkage in V3 after document and baseline traceability are stable.

The forecast is a planning aid. It must show its baseline, assumptions and recalculation state. Commercial pricing, delivery guarantees and operational service levels are separate product decisions.

27 Decisions glossary and handoff checklist

Product decisions to close

Decision	Required resolution
Persistence and retention	Which source, findings and decisions are saved; for how long; how deletion and recovery work.
External AI policy	Who authorizes processing, which documents are restricted and how exceptions are recorded.
Workspace permissions	Which users can upload, review, approve, export and manage a project.
Supported inputs and previews	Format and OCR roadmap; which UI patterns the first Playground supports.
Estimation and commercial limits	Effort method, capacity model, operational budget, pricing and validated service targets.

Terms used in this document

Source evidence is a passage or table relationship from a specific document version. A finding is an interpretation linked to that evidence. A user answer has separate provenance. A baseline is an approved set of findings and scope decisions. AI-reviewed means a model review accepted a draft; human-approved means an authorized person made a recorded decision. A forecast is a calculated delivery range based on explicit inputs.

Handoff checklist

- Confirm the baseline and the difference between live, planned and deferred work.
- Resolve processing policy, persistence and workspace authorization before enterprise onboarding.
- Keep every acceptance check linked to its requirement and source version.
- Validate the complete browser journey and real representative documents under the permitted processing policy.
- Preserve existing backend data and integrations during any future migration.

Reference baseline

Product decisions derive from the Rulebook planning discussions and the original nine-group roadmap. Current behavior is grounded in the source-first understanding notes, Group 3 facts notes, Group 4 scope notes, reader implementation and protected Vercel release verified on 30 October 2026. The current entry is https://pragi-rulebook.vercel.app/.

28 Current API and identity contract

This appendix describes the current private-demo request and response boundary. Persistent project and review endpoints require a separate implementation contract; no unimplemented URL is implied here.

Request identity and source

Field	Current constraint
documentId	A session identifier used to bind a response to the selected document. It is not an enterprise workspace authorization token.
consent	Must be true for an AI POST. The server rejects requests without explicit consent.
pages	All pages in order, including blank pages; each has an authoritative page number and extracted text. Validated warnings and table metadata may be attached.
Source unit and item IDs	Assigned by the application. A review must refer to the explicit item identity, not an inferred list position.

Response requirements

The understanding response carries purpose, the short understanding, role requirements, facts and source coverage. The scope response carries areas, scope items, source dispositions, original sources, warnings, counts and safe run diagnostics. Both are bound to documentId and marked as needing review. The browser rebuilds source identity before accepting them.

HTTP status	Meaning	Required handling
400	Invalid JSON schema or missing consent	Correct the request; do not retry blindly.
403	Wrong private host or request origin	Use the protected deployment and the proper origin.
413	Source or inventory limit exceeded	Keep the reader and split the document.
422	Insufficient readable scope content	Request a readable source; no AI call is made.
429	Provider rate limit	Preserve completed results and allow later retry.
502 or 503	Incomplete analysis or unavailable configuration	Show a safe error; retain local reading.

The live service rejects client model overrides and browser API keys. AI POST results use no-store caching. Health metadata exposes configuration and version flags, never a secret or evidence that an analysis has succeeded.